



TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CIVICS GROUP

H2 BIOLOGY

9744/01

Paper 1 Multiple Choice

23 September 2022

1 hour

Additional material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the Multiple Choice Answer Sheet.

Do not use staples, paper clips, glue or correction fluid/tape.

There are **30** questions in this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

You may keep this booklet after the exam.

QUESTION 1

Which of the following observations does **not** support the Cell Theory?

- A. Glucose molecules that enter a cell are quickly oxidized to pyruvate molecules.
- B. Plant cells placed in a hypotonic solution grow rapidly in size.
- C. One cancer cell rapidly forms a cluster of cells.
- D. The amount of nuclear DNA in a cell doubles during S phase of the cell cycle.

QUESTION 2

Oxygen molecules produced during photosynthesis in a plant cell are often used directly by the same plant cell for cellular respiration.

How many phospholipid **layers** must an oxygen molecule produced during photosynthesis cross in order to reach its functioning destination in cellular respiration?

- A. 4
- B. 5
- C. 8
- D. 10

QUESTION 3

The features of the cell membrane are described below.

1. phospholipid molecules move laterally within each phospholipid layer
2. the cell surface membrane can extend around foreign particle and engulf it by phagocytosis
3. integral proteins are scattered within the phospholipid bilayer
4. both saturated and unsaturated phospholipid molecules are present in both phospholipid layers

Which features are described by the fluid mosaic model of cell membranes?

- A. 1 and 4 only
- B. 1 and 3 only
- C. 2 and 4 only
- D. 1, 2, 3 and 4

QUESTION 4

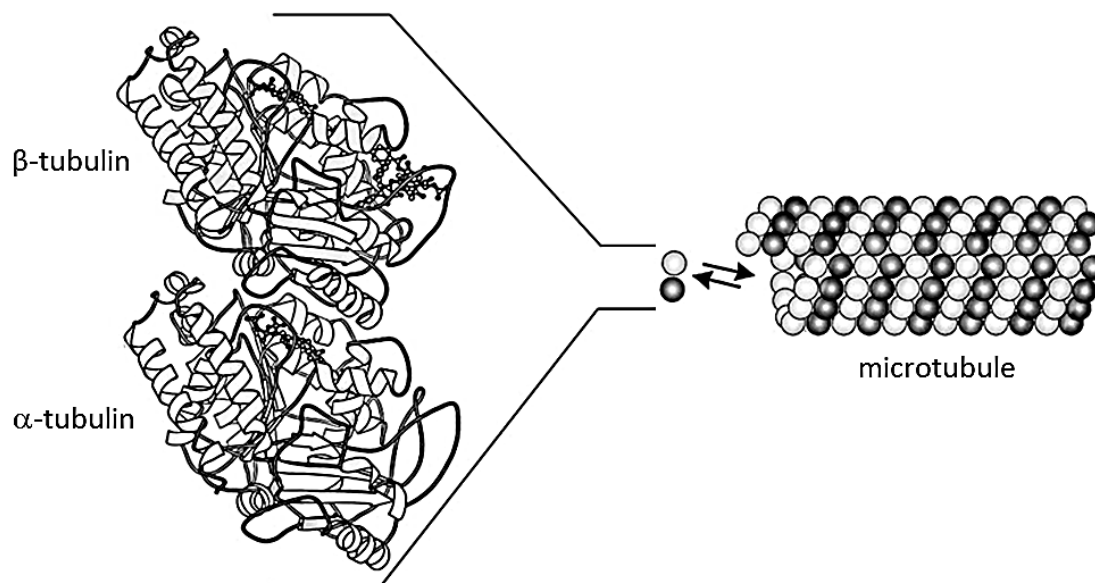
A typical human red blood cell contains 2.5×10^8 haemoglobin molecules.

Which of the following is true of a human red blood cell?

	number of α -globin chains present in a human red blood cell	maximum number of oxygen molecules a human red blood cell can carry
A.	5.0×10^8	2.5×10^8
B.	5.0×10^8	1.0×10^9
C.	1.0×10^9	2.5×10^8
D.	1.0×10^9	1.0×10^9

QUESTION 5

Microtubules are assembled from a heterodimeric protein called tubulin, as shown in the diagram below.



Which of the following explains why tubulin is considered as a monomer as well as a polymer?

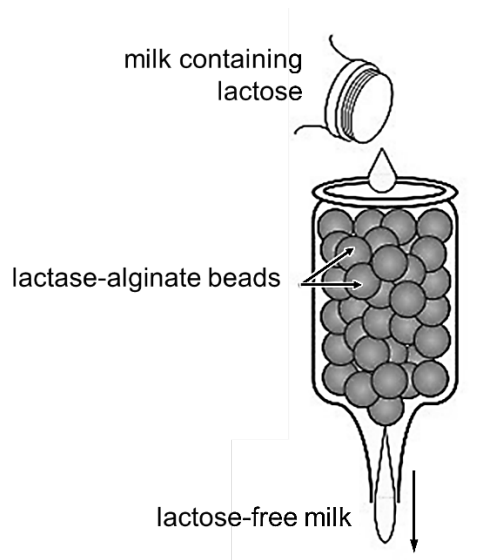
	why tubulin is a monomer	why tubulin is a polymer
A.	It is made up of many amino acids.	It is the building block of microtubule.
B.	It is the building block of microtubule.	It is made up of many amino acids.
C.	It is a quaternary protein.	It is a macromolecule.
D.	It is made up of two different subunits.	It has many α -helices and β -pleated sheets.

QUESTION 6

Enzymes, such as lactase, are often immobilized for use in the production of lactose-free milk.

Lactase is immobilized by encasing it in a gel-like matrix known as alginate beads. The beads vary in size. Small beads are usually selected and placed into a glass column, as shown in the diagram.

As milk that contains lactose passes through the column and comes into contact with the beads, lactose is broken down into its monosaccharides, glucose and galactose.



Which is **not** an advantage of using small beads compared to using large beads?

- A. Small beads have larger surface area to volume ratio.
- B. Small beads allow more time for lactase to be exposed to milk.
- C. Small beads allow more lactase to be exposed to milk at any one time.
- D. Small beads accelerate the passage of milk through the column.

QUESTION 7

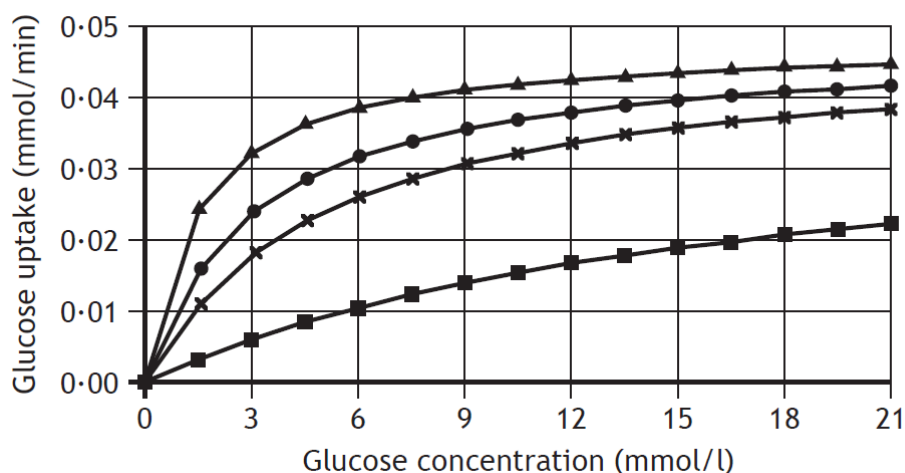
The diffusion of blood glucose across the cell surface membrane of mammalian cells is facilitated by a family of related proteins called glucose transporters (GLUT).

A study measured the changes in rates of glucose uptake by different forms of GLUT as the concentration of glucose is increased. Different forms of GLUT are found in different tissues.

For each type of GLUT, the rate of transport levels off to a maximum value that is termed $V_{\max}$.

The glucose concentration at which the rate of transport is **half** $V_{\max}$ is defined as the K_M of the transporter. K_M is an indication of the affinity of the GLUT for glucose.

The K_M for the four forms of GLUT and the effect of increasing glucose concentration of the rate of glucose uptake are shown below.



Key

▲▲ GLUT3 ($K_M = 1.4$ mmol/l) ●● GLUT1 ($K_M = 3.0$ mmol/l)
 ✕✕ GLUT4 ($K_M = 5.0$ mmol/l) ■■ GLUT2 ($K_M = 17.0$ mmol/l)

Which statement can be concluded from the study?

- A. GLUT3 has the lowest affinity for glucose.
- B. The $V_{\max}$ for GLUT2 is 0.04 mmol/min.
- C. GLUT1, GLUT3 and GLUT4 transport glucose via facilitated diffusion, while GLUT2 transport glucose via active transport.
- D. As glucose concentration is further increased beyond 21 mmol/l, all the four forms of GLUT will eventually reach the same $V_{\max}$.

QUESTION 8

Which of the following describes the fate of the products of the photolysis of water during photosynthesis?

	products of photolysis of water	fate
A.	ATP NADPH oxygen molecule	involved in the reduction stage of Calvin cycle involved in the reduction stage of Calvin cycle serves as the final electron acceptor during respiration
B.	electrons protons hydrogen molecule	replace the electrons lost in photosystem II contribute to the proton gradient in the thylakoid space diffuses out of the plant cell
C.	carbon dioxide hydroxide ions oxygen molecule	combines with RuBP in the Calvin cycle combine with protons to regenerate water serves as the final electron acceptor during respiration
D.	electrons protons oxygen molecule	replace the electrons lost in photosystem II contribute to the proton gradient in the thylakoid space serves as the final electron acceptor during respiration

QUESTION 9

Removal of the source of carbon dioxide from photosynthetic plant cells results in rapid changes in the concentration of ATP, ribulose biphosphate and glycerate-3-phosphate.

Which of the following shows the correct changes in concentration?

	ATP	ribulose biphosphate	glycerate-3-phosphate
A.	decreases	decreases	increases
B.	decreases	increases	decreases
C.	increases	increases	decreases
D.	increases	decreases	increases

QUESTION 10

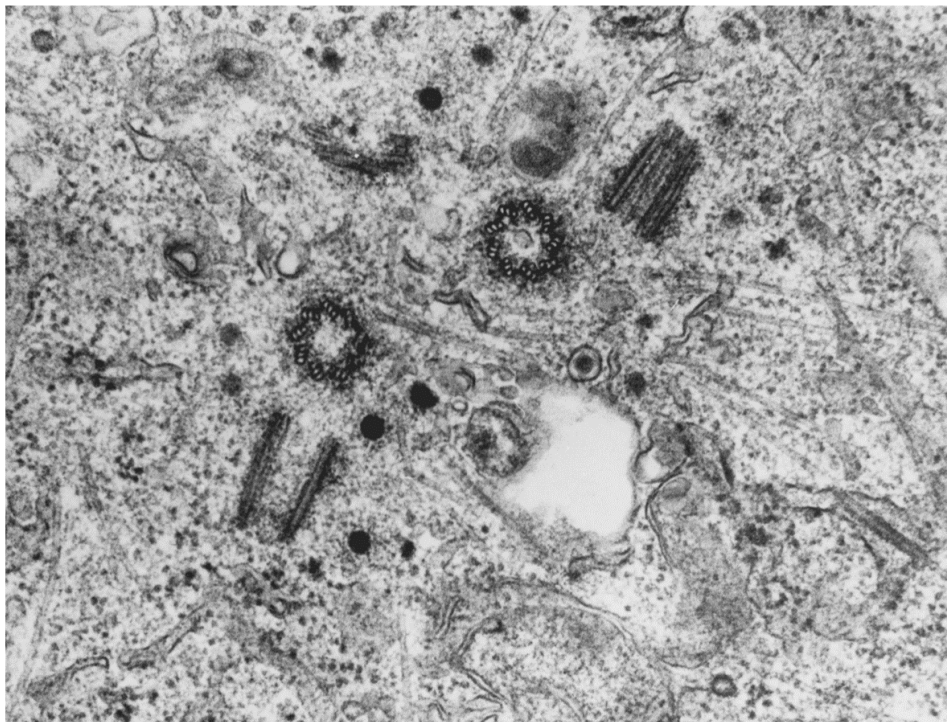
Which statements correctly describe cellular respiration?

1. Two turns of the Krebs cycle are required to oxidize one molecule of glucose.
2. Four molecules of carbon dioxide are generated for every molecule of acetyl-coA introduced into the Krebs cycle.
3. During aerobic respiration, glucose produces pyruvate, carbon dioxide and ATP in the cytoplasm of a muscle cell.
4. Aerobic respiration produces about 19 times the amount of ATP produced in anaerobic respiration per glucose molecule.

A. 1 only B. 1 and 4 only C. 2 and 4 D. 1, 3 and 4

QUESTION 11

The electron micrograph shows part of a stem cell that is undergoing mitosis.



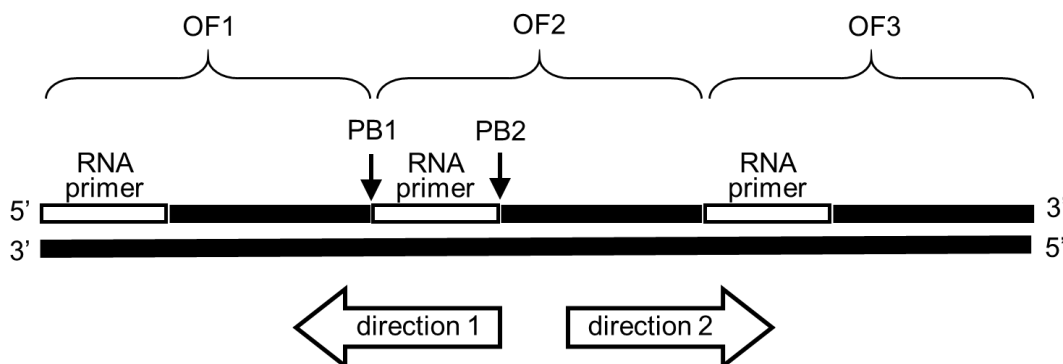
1 μ m

Which mitotic phase is the stem cell undergoing?

A. prophase B. metaphase C. anaphase D. telophase

QUESTION 12

The diagram shows a DNA template with the lagging strand prior to the removal of the RNA primers.



Which row correctly shows the events taking place during the synthesis of the lagging strand?

	last Okazaki fragment synthesised	phosphodiester bond formation catalysed by...		polymerization of the lagging strand	polymerization of the RNA primer
		DNA ligase	DNA polymerase		
A.	OF3	PB2	PB1	direction 1	direction 1
B.	OF1	PB2	PB1	direction 2	direction 2
C.	OF3	PB1	PB2	direction 1	direction 2
D.	OF1	PB1	PB2	direction 1	direction 2

QUESTION 13

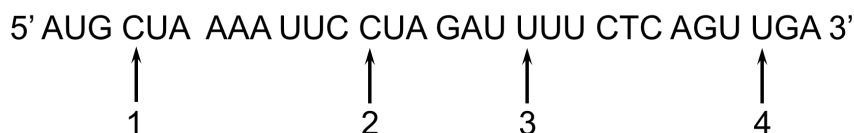
What do DNA replication, transcription and translation have in common?

1. formation of polymers
2. breaking and forming of hydrogen bonds
3. involvement of only one enzyme
4. reading of the template strand from the 3' to 5' direction

A. 1 and 2 B. 1 and 3 C. 2 and 3 D. 4 only

QUESTION 14

An mRNA contains the following codons. Four of the nucleotides are labelled 1 to 4.

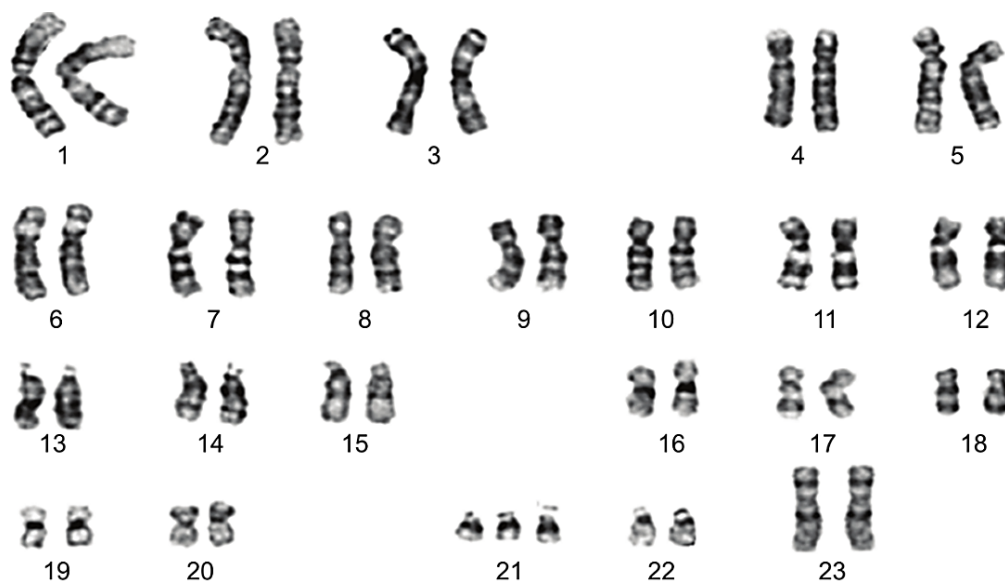


Which nucleotides, when deleted, result in a shortened polypeptide chain?

A. 1 and 3 B. 2 and 4 C. 3 and 4 D. 1 and 2

QUESTION 15

The diagram shows a karyotype (an individual's collection of chromosomes) taken from a patient suffering from a disease.



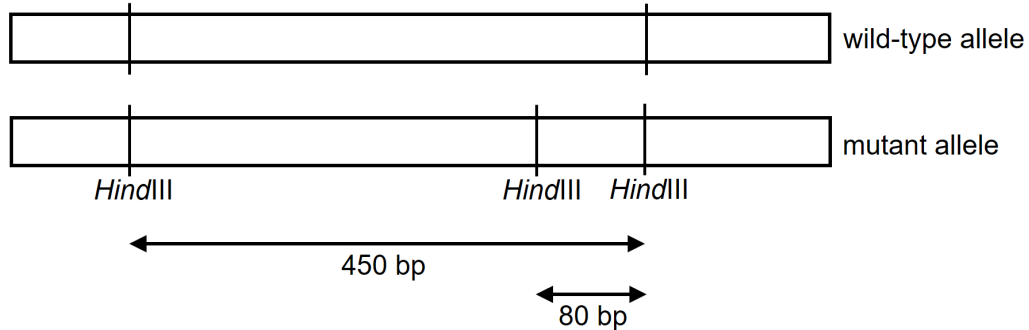
What would be the most accurate description of the cause of the disease and the phenotypes of the patient?

	cause of the disease	phenotypes of the patient
A.	aneuploidy	female, sickle-shaped red blood cells
B.	trisomy	female, mental retardation and up-slanting eyes
C.	polyploidy	male, sickle-shaped red blood cells
D.	chromosomal aberration	male, mental retardation and up-slanting eyes

QUESTION 16

A patient is suspected to be heterozygous for a gene (650bp) associated with mild neurological disorder. The mutated form of the gene arises as a result of a single nucleotide substitution that generates an extra *Hind*III restriction site.

The *Hind*III restriction sites on the wild-type (normal) allele and mutant allele are shown in the diagram.



You collected a blood sample from the patient and extracted the **total genome** from the white blood cells.

Which molecular techniques should be used to ascertain the genotype of this patient?

	polymerase chain reaction	restriction digestion with <i>Hind</i> III enzyme	nucleic acid gel electrophoresis	Southern blotting	nucleic acid hybridization
A.	x	✓	✓	✓	✓
B.	✓	x	✓	x	x
C.	x	✓	✓	x	x
D.	✓	✓	x	✓	✓

key

✓ = used

x = not used

QUESTION 17

A ribonucleoprotein is a complex of ribonucleic acid (RNA) and proteins.

Which are examples of ribonucleoprotein?

1. influenza virus
2. T4 bacteriophage
3. RNA primase
4. spliceosome
5. telomerase
6. ribosome

A. 1, 2 and 3

B. 1, 3, and 6

C. 3, 5 and 6

D. 4, 5 and 6

QUESTION 18

The discovery of introns and splicing in the 1970s led to two theories of their origin that became known as Introns Early and Introns Late.

- The Introns Early theory proposed that introns were present in the common ancestor of prokaryotes and eukaryotes. Introns then suffered different fates in the different lineages: they were progressively lost in prokaryotic lineages but persisted in eukaryotic lineages.
- The Introns Late theory proposed that introns first appeared in eukaryotes and have been inserted into protein-coding genes continuously throughout the evolution of eukaryotes.

Three observations were made regarding introns.

1. In many homologous genes found in animals and plants, introns intervene between exons at the same positions, suggesting that these introns were in place prior to the plant–animal evolutionary split.
2. A small number of prokaryotic genes contain what is known as ‘self-splicing introns’. These introns, when transcribed into RNA, are ribozymes that catalyse their own excision.
3. Spliceosome genes are absent in prokaryotes but present in eukaryotes. Spliceosome genes have evolved in eukaryotes as a means to remove introns in mRNA transcribed from nuclear genes.

Which theory is supported by these observations?

	Introns Early	Introns Late
A.	1 and 3	2
B.	2	1 and 3
C.	3	1 and 2
D.	1 and 2	3

QUESTION 19

The worrying reports of antibiotic resistance and limited new antibiotic discoveries have fuelled innovation in other research fields and led to a revitalization of bacteriophage (phage) studies in the Western world.

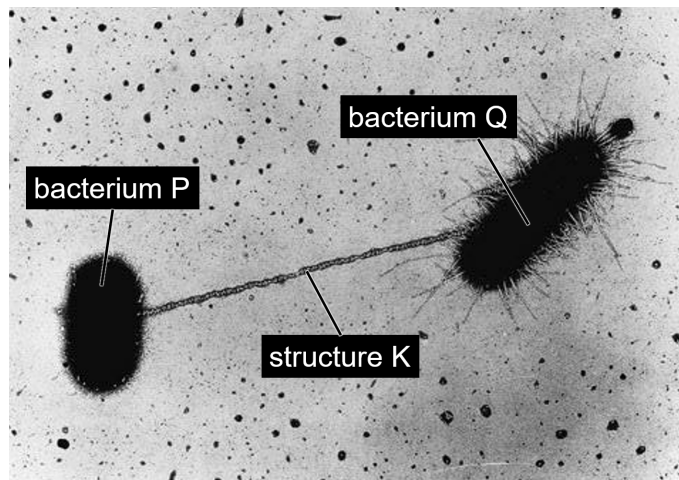
Phage therapy mainly utilizes lytic phages to kill their respective bacterial hosts, while leaving human cells intact. Phage therapy is rapidly evolving and has saved thousands of lives in clinical trials and emergencies.

What would compel doctors to use phage therapy over antibiotic therapy?

- A. Bacteriophage is highly specific; they kill only the bacteria that they can recognize. In contrast, antibiotics are non-discriminatory and kill a broad range of bacterial cells.
- B. Phage therapy is effective in treating bacterial infections that do not respond to antibiotics.
- C. Only one dose of bacteriophage is required, since bacteriophage multiply and increase in number by themselves during treatment. In contrast, antibiotic therapy requires multiple doses over a few days.
- D. Phages are significantly safer and better tolerated, as they replicate only in bacterial cells but cannot infect mammalian cells. In contrast, antibiotics can result in side effects such as allergic reactions.

QUESTION 20

The micrograph shows a process occurring between two bacterial cells.



Five statements concerning the process are made:

- Structure K is a membrane extension connecting between bacterium P to bacterium Q and it facilitates the transfer of genetic material between the two bacteria.
- A polynucleotide is transferred, 3' end first, from bacterium Q to bacterium P.
- Structure K is made up of protein subunits that are coded for by the genes carried on the chromosomal DNA in bacterium Q.
- At the end of the process, bacterium P will be genetically identical to bacterium Q.
- DNA replication is taking place in bacterium P but not in bacterium Q.

How many statement(s) is/are correct?

- A. 0
- B. 1
- C. 3
- D. 5

QUESTION 21

Ampicillin-sensitive *E. coli* cells were subjected to heat-shock transformation to take up plasmid DNA that contain the ampicillin-resistance gene.

Successfully transformed cells were selected and allowed to divide for several generations in a nutrient medium without ampicillin. It was observed that some of the *E. coli* cells were no longer resistant to ampicillin.

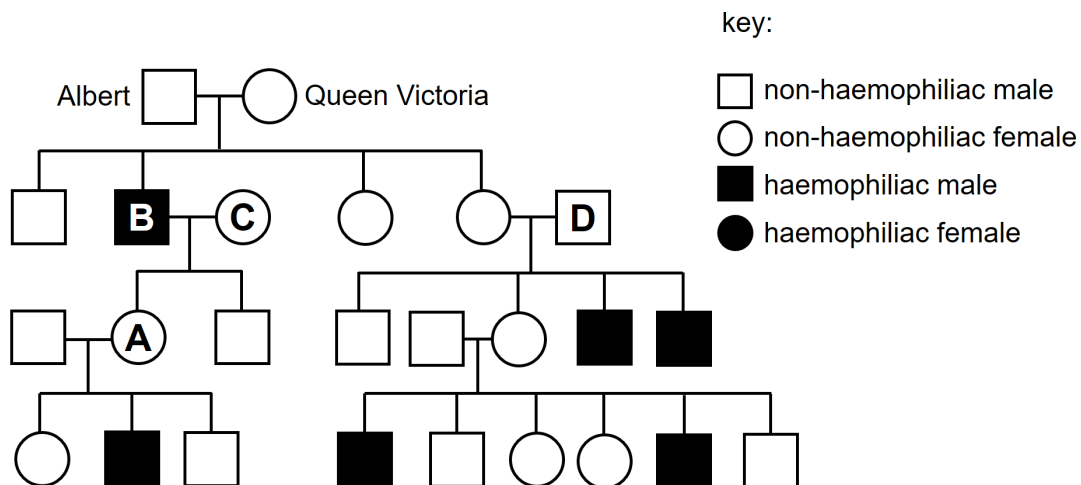
What is the most likely explanation for the phenomenon?

- A. Error during DNA replication causes a loss-of-function mutation to the ampicillin-resistance gene.
- B. Incorporation of the plasmid DNA into the heterochromatin region of the bacterial genome renders the ampicillin-resistance gene inactive.
- C. Restriction enzymes present in *E. coli* cells recognize specific sequences on the plasmid DNA and hydrolyse it into fragments.
- D. Unequal distribution of plasmid during binary fission results in some daughter cells without plasmids.

QUESTION 22

The pedigree chart shows the inheritance of haemophilia (a sex-linked disease) in some of the descendants of Queen Victoria.

Which letter represents a descendant certain to be heterozygous?



QUESTION 23

Two genetic crosses in a species of flies were performed. In both crosses, the parents were pure breeding for eye colour and body colour.

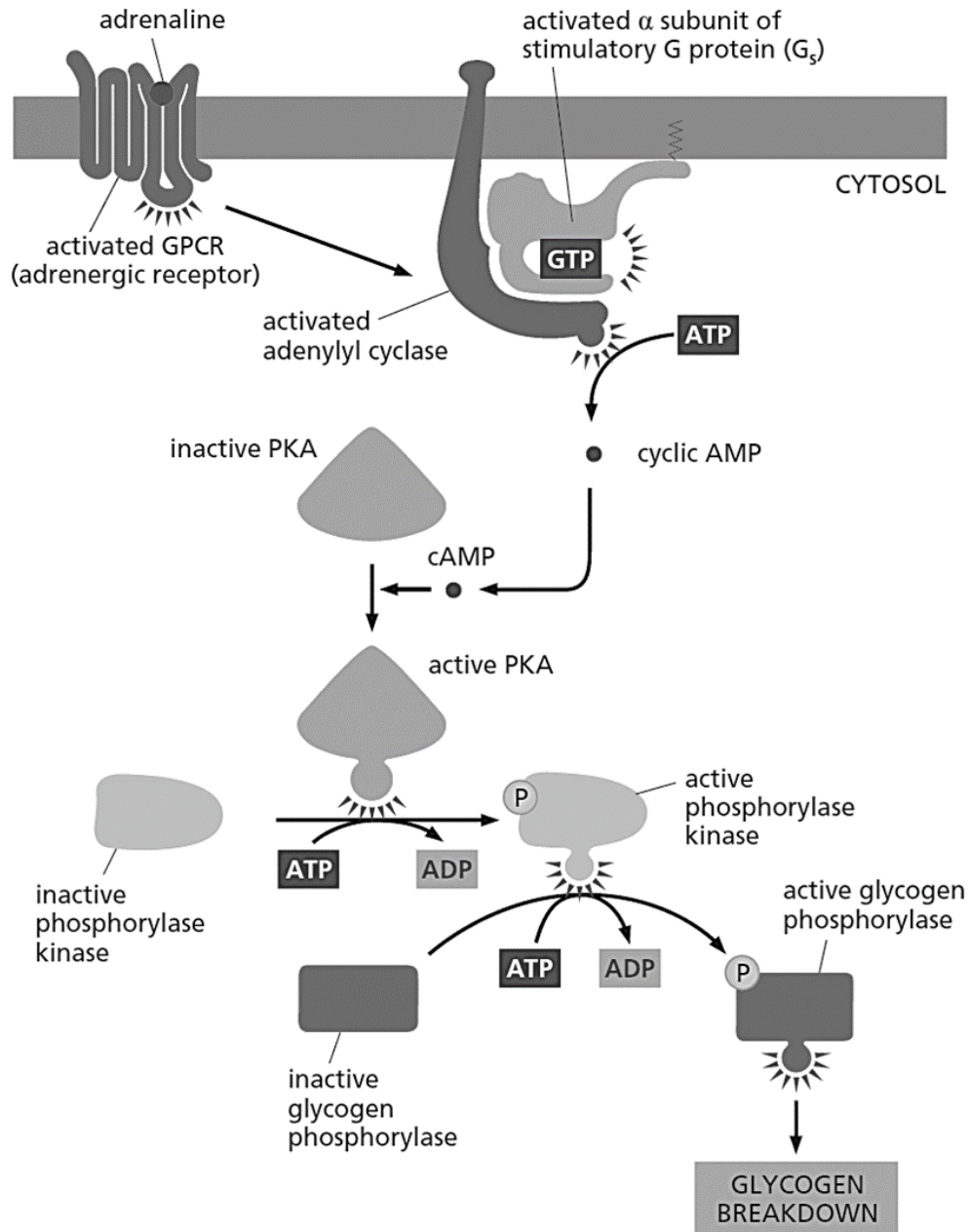
	male	female	offspring
cross 1	red-eyed, black body	white-eyed, grey body	all females are red-eyed, all males are white-eyed, both sexes have black body
cross 2	white-eyed, grey body	red-eyed, black body	both sexes are red-eyed, both sexes have black body

What can be concluded from the results of the two crosses?

	inheritance of eye colour	inheritance of body colour
A.	sex-linked	autosomal
B.	autosomal	sex-linked
C.	sex-linked	sex-linked
D.	autosomal	autosomal

QUESTION 24

The diagram shows the series of molecular events that take place when the hormone adrenaline binds to an adrenergic receptor, a G-protein coupled receptor, on the cell surface membrane of a skeletal muscle cell, leading to glycogen breakdown.



How many amplification steps are there in this transduction pathway?

- A. 1 B. 2 C. 4 D. 6

QUESTION 25

In one of the scientific journals that Charles Darwin published in 1839, he wrote:

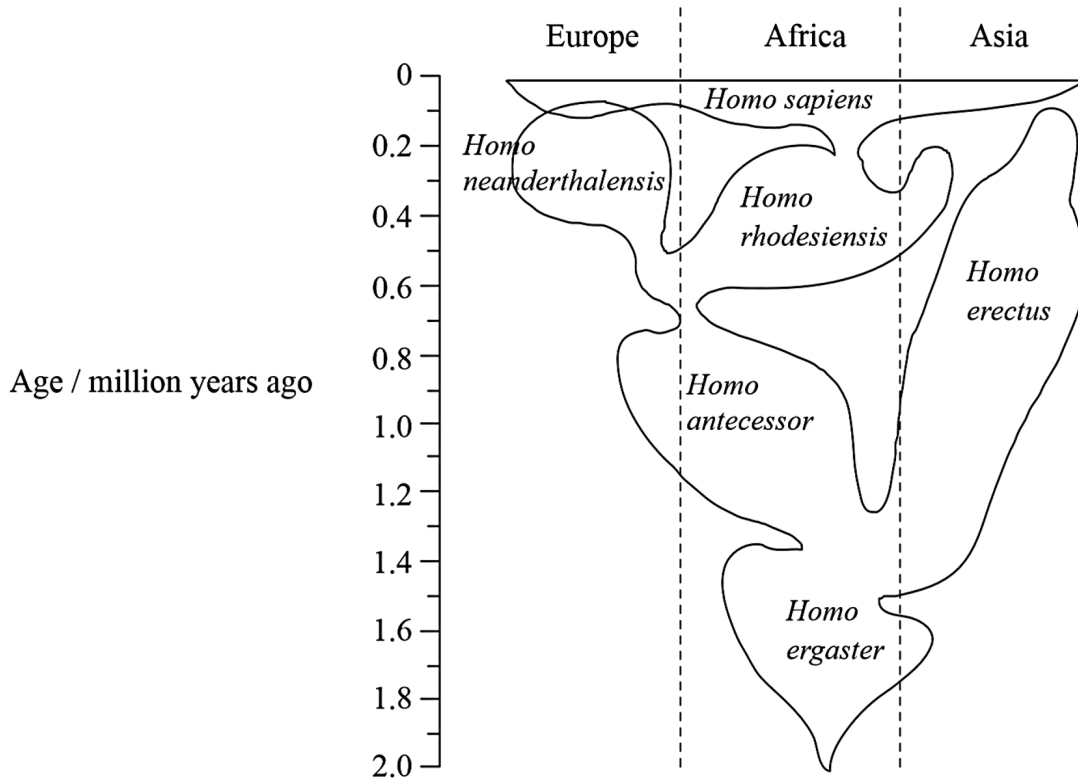
"I have stated, that in the thirteen species of ground-finches [in the Galápagos Island], a nearly perfect gradation may be traced, from a beak extraordinarily thick, to one so fine, that it may be compared to that of a warbler. I very much suspect that certain members of the series are confined to different islands; therefore, if the collection had been made on any one island, it would not have presented so perfect a gradation. It is clear, that if several islands have each their peculiar species of the same genera, when these are placed together, they will have a wide range of character."

What does the above quote illustrate?

	type of speciation	type of variation	type of evolution	type of structure
A.	sympatric	continuous	divergent	analogous
B.	allopatric	discontinuous	convergent	analogous
C.	allopatric	continuous	divergent	homologous
D.	allopatric	discontinuous	convergent	homologous

QUESTION 26

Recently discovered fossilized partial skulls from Ethiopia have raised new questions about early human evolution. This has led to different theories about the origins of *Homo sapiens*. One of these theories is illustrated in the diagram below.



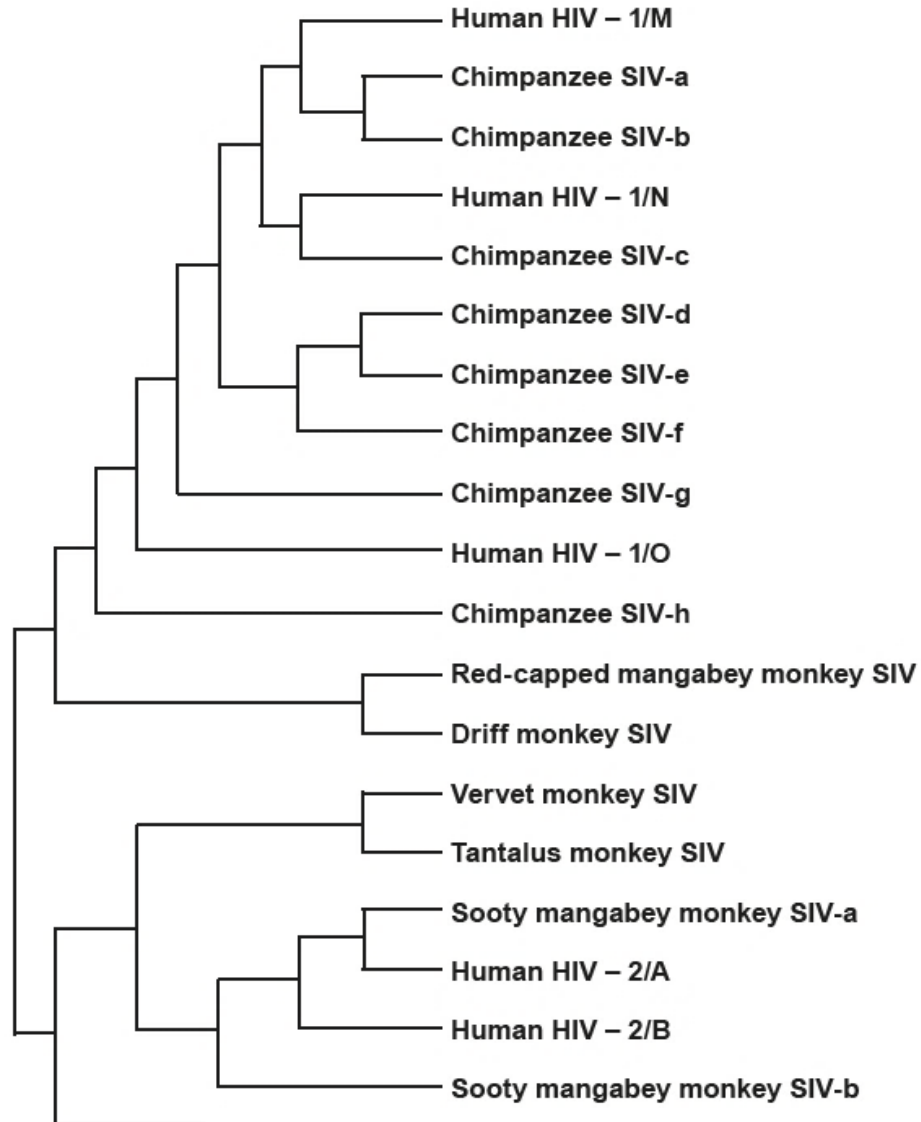
What can be concluded from the diagram?

	species that shows the greatest geographical distribution	species that likely provides the most fossil evidence
A.	<i>Homo erectus</i>	<i>Homo erectus</i>
B.	<i>Homo erectus</i>	<i>Homo antecessor</i>
C.	<i>Homo sapien</i>	<i>Homo erectus</i>
D.	<i>Homo sapien</i>	<i>Homo ergaster</i>

QUESTION 27

There are two main forms of the Human Immunodeficiency Virus: HIV-1, and the more virulent HIV-2. It is believed that HIV-1 and HIV-2 appeared due to mutations of a virus (SIV) found in monkeys and chimpanzees. These mutations have enabled them to infect humans.

Scientists have analysed the molecular similarities between HIV from infected humans and similar viruses which are found in monkeys and chimpanzees. The phylogenetic tree shows the possible evolutionary history of HIV.



Key: SIV = Simian Immunodeficiency Virus.

HIV = Human Immunodeficiency Virus.

What can be concluded regarding the evolution of both forms of human HIV?

- A. HIV-2 evolved earlier than HIV-1.
- B. HIV-2 and chimp SIV diverged from a common ancestor.
- C. HIV-1 evolved thrice and HIV-2 evolved twice.
- D. HIV-1 and HIV-2 evolved from a common HIV ancestor.

QUESTION 28

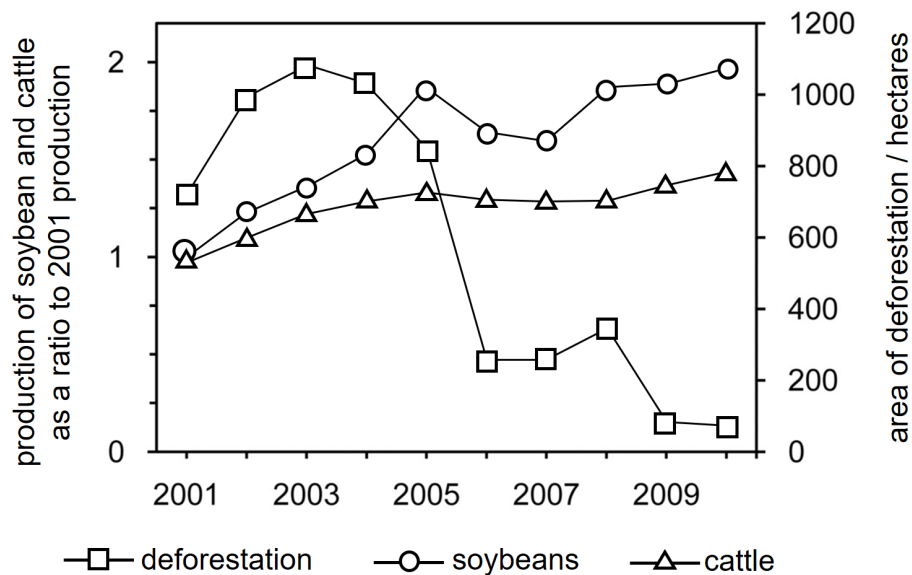
Which row correctly shows the mechanism of action of the antibiotic penicillin?

	cellular process interfered	organelle or enzyme inhibited	effect on the bacterial cell
A.	protein synthesis (translation)	70S ribosomes	bactericidal
B.	protein synthesis (translation)	80S ribosomes	bacteriostatic
C.	peptidoglycan cell wall synthesis	transpeptidase	bactericidal
D.	peptidoglycan cell wall synthesis	cellulose synthase	bacteriostatic

QUESTION 29

Anthropogenic activities such as deforestation and agricultures are believed to be the major contributors to enhanced greenhouse effect.

The graph shows the annual trends in the area deforested and the production of soybeans and cattle in the state of Mato Grosso in Brazil in the period from 2001 to 2010.



Which of the following statements can be deduced from the information in this graph?

1. Soybean production contributed more to deforestation than cattle production.
2. Many animals lost their habitats in the forest between 2001 and 2002.
3. Carbon sinks declined between 2003 and 2006 and between 2008 and 2009.
4. There has been an increase in consumers' demand for cattle's milk and meat over the period of study.

A. 1 and 2

B. 2 and 3

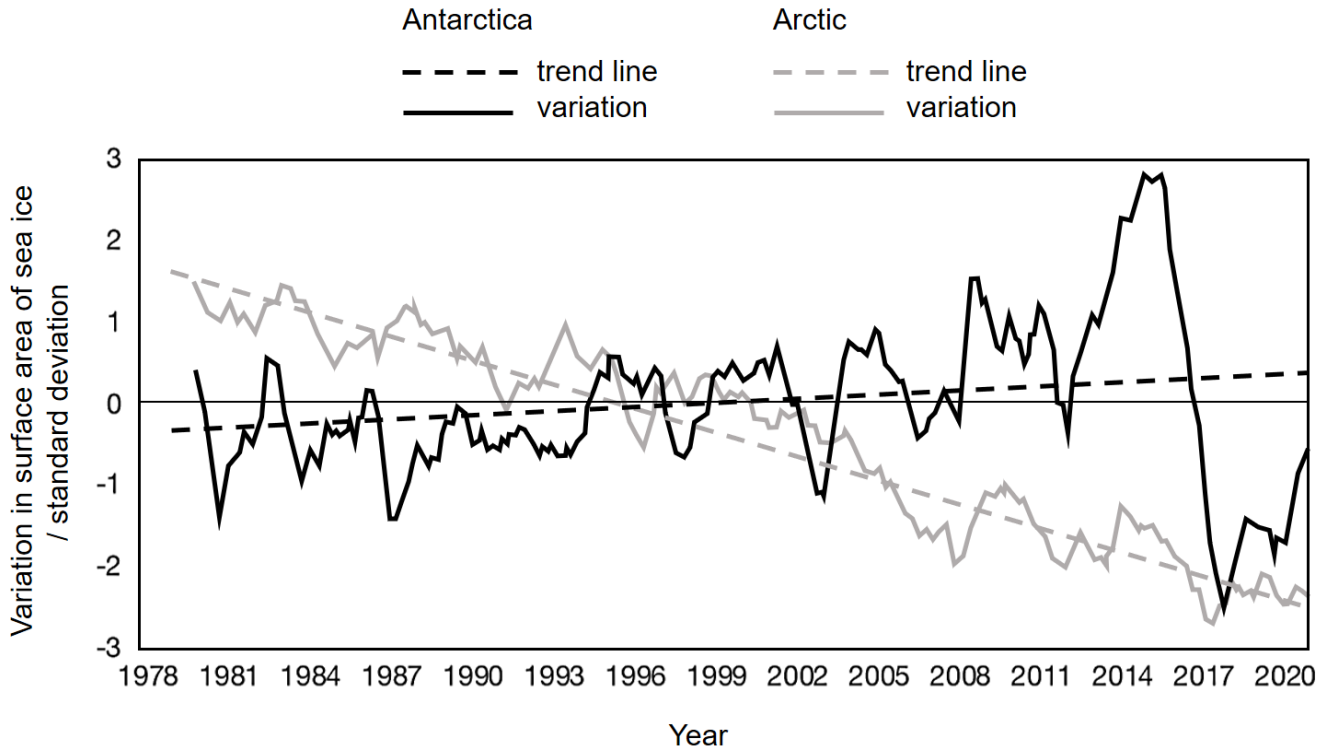
C. 2 and 4

D. 1 and 4

QUESTION 30

Global warming has changed both the thickness and surface area of sea ice of the Arctic Ocean as well as the Southern Ocean that surrounds Antarctica. Sea ice is highly sensitive to changes in temperature.

Scientists have calculated a long-term mean (from 1981-2010) for the surface area of sea ice in the Arctic and in the Southern Ocean around Antarctica. This mean value is used as a reference for changes in surface area of sea ice. The graph shows the variations (standard deviation) from this mean (zero line) over a period of time.



What can be possibly deduced from the graph?

1. The rate of change in surface area of sea ice is greater for the Arctic than for the Antarctica, but there are greater fluctuations in the surface area of sea ice in the Antarctica than in the Arctic.
2. Melting of sea ice is expected with global warming, but the general increase in variation of surface area of sea ice in the Antarctica Ocean is evident that global warming has complex effects and does not affect all areas in the same way.
3. Penguins in the Antarctica Ocean faced immense hunting by marine predators from 2014 to 2017 due to a drastic loss of dry spots for hiding.
4. Polar bears in the Arctic are in danger of a drastic decrease in population from the retreating sea ice that serves both as their habitat and their hunting ground for seals.

A. 1 only B. 2 and 3 only C. 3 and 4 only D. 1, 2, 3 and 4

☺ **END OF PAPER 1** ☺